Set A. Macroeconomics Case Study Practice — Short-Answer Questions

Instructions to students. This booklet contains four case studies spanning contemporary macroeconomic problems: Singapore's growth deceleration and the MAS policy mix; the deepening US–China trade and technology rivalry and its supply-chain implications; Japan's long-running deflation and the BOJ's exit from yield-curve control; and Germany's twin energy and fiscal-debt-brake crisis. Each case study contains a data table and an extract, followed by short-answer questions ranging from 2 to 6 marks.

Use the mark allocation as a guide to depth: 2-mark questions require a precise definition or identification with reference to the data; 4-mark questions require a developed two-step explanation; 6-mark questions require a fully developed analytical response supported by an AD/AS diagram or relevant data. Write your answers in the lined spaces provided.

Section Case Study 1. Singapore — Growth Deceleration and the MAS Policy Mix

Table 1.1: Selected Singapore Macroeconomic Indicators, 2020–2024

Indicator	2020	2021	2022	2023	2024
Real GDP Growth (%)	−3.9	9.7	3.8	1.1	4.0
MAS Core Inflation (%)	−0.2	0.9	4.1	4.2	2.8
Headline CPI Inflation (%)	−0.2	2.3	6.1	4.8	2.4
Resident Unemployment Rate (%)	4.1	3.5	2.9	2.8	2.7
S$NEER Appreciation Rate (%, slope est.)	0.0	0.5	2.0	1.5	1.5
Productivity Growth (%, value-added per hour)	−2.1	5.4	0.4	−1.9	0.8
Goods + Services Trade-to-GDP Ratio (%)	295	318	339	311	319

Source: Adapted from Monetary Authority of Singapore Macroeconomic Review, Department of Statistics Singapore, and Ministry of Manpower.

Extract 1.1: MAS pauses tightening as core inflation moderates, but productivity gap persists

The Monetary Authority of Singapore (MAS) held the slope and centre of the Singapore dollar nominal effective exchange rate (S$NEER) policy band unchanged at its January 2024 and April 2024 meetings, formally ending the most aggressive monetary policy tightening cycle in Singapore's history. Between October 2021 and October 2022, MAS executed five consecutive tightening moves, including two unscheduled off-cycle decisions, in response to the surge in imported inflation triggered by the Russia-Ukraine conflict and pandemic-era supply chain disruptions.

The April 2024 Monetary Policy Statement noted that MAS Core Inflation, which excludes accommodation and private transport costs and serves as the primary measure of underlying inflationary pressure, had moderated from a peak of 5.5 per cent in early 2023 to 2.8 per cent in 2024. Headline CPI inflation, which has fallen even more rapidly, was buoyed in 2023 by domestic services inflation and the one-percentage-point GST increase to 9 per cent in January

2024. "The current appreciating policy stance remains appropriate," MAS said, "but further tightening is not warranted given the projected continued moderation in imported inflation."

However, the same Macroeconomic Review flagged a structural concern: productivity growth, which had averaged 2.3 per cent per annum in the decade before the pandemic, had collapsed to near zero. "A persistent productivity slowdown would lower Singapore's potential output and require a corresponding adjustment in the appreciation path of the S$NEER," the report warned. The Ministry of Trade and Industry projected real GDP growth of 1 to 3 per cent for 2024, below the long-run trend of 2 to 3 per cent. Economists at major Singapore banks argued that the slowdown reflected not only weaker external demand from China and the eurozone but also a maturing economy where labour-input and capital-deepening contributions to growth had largely been exhausted.

Source: Adapted from The Business Times, 12 April 2024 and MAS Macroeconomic Review, April 2024.

Question 1 **[2 marks]**

With reference to Table 1.1, compare the trend in MAS Core Inflation with the trend in Headline CPI Inflation between 2020 and 2024.

Question 2 **[2 marks]**

Define "potential output" as used in Extract 1.1, and identify one piece of evidence from the extract that suggests Singapore's potential output growth has slowed.

Question 3 **[4 marks]**

With reference to Extract 1.1 and Table 1.1, explain why MAS uses Core Inflation, rather than Headline CPI Inflation, as the primary indicator of underlying inflationary pressure.

Question 4 **[4 marks]**

Explain one reason why Singapore's trade-to-GDP ratio of over 300 per cent makes the exchange rate an effective instrument for managing inflation, and one reason why it limits the effectiveness of fiscal policy.

Question 5 **[6 marks]**

With the aid of an AD/AS diagram, explain how a persistent slowdown in productivity growth, as described in Extract 1.1, could reduce Singapore's long-run potential output and raise underlying inflationary pressure.

Question 6 **[6 marks]**

Using evidence from Extract 1.1 and Table 1.1, explain why MAS decided to pause its tightening cycle in 2024 rather than ease the appreciation path of the S$NEER.

Section Case Study 2. United States and China — Trade Tensions and Supply-Chain Realignment

Table 2.1: Selected Bilateral and Macroeconomic Indicators, 2018–2023

Indicator	2018	2019	2020	2021	2022	2023
US Imports from China (US$ billion)	540	452	435	506	536	427
US Imports from Vietnam (US$ billion)	49	67	79	102	127	114
US Imports from Mexico (US$ billion)	346	358	325	384	455	476
US Average Tariff on Chinese Imports (%)	3.1	19.3	19.3	19.3	19.3	19.3
China FDI Inflows (US$ billion)	138	141	149	181	189	33
US Real GDP Growth (%)	3.0	2.6	−2.2	5.8	1.9	2.5
China Real GDP Growth (%)	6.7	6.0	2.2	8.4	3.0	5.2

Source: Adapted from US Census Bureau, Peterson Institute for International Economics, and UNCTAD World Investment Report 2023.

Extract 2.1: The decoupling that wasn't, and the re-routing that was

In 2018, the United States imposed tariffs averaging 19.3 per cent on roughly two-thirds of imports from China, citing concerns over intellectual property theft, forced technology transfer, and the bilateral trade deficit. The Biden administration retained the Trump-era tariffs and added export controls on advanced semiconductors, with the CHIPS and Science Act of 2022 providing US$52 billion in subsidies to onshore semiconductor manufacturing.

The headline impact has been striking: in 2023, US goods imports from China fell to US$427 billion, the lowest since 2010, and China ceded its long-held position as the largest source of US imports to Mexico. Vietnam's exports to the US more than doubled between 2018 and 2022. Yet many economists argue that "decoupling" is a misnomer. A study by the Federal Reserve Bank of New York found that Chinese inputs continue to dominate the supply chains of Vietnamese and Mexican exports to the US, with up to 40 per cent of the value added in some product categories still traceable to Chinese suppliers. "What we are observing is not decoupling but re-routing," the study concluded.

For consumers, the empirical evidence on tariff incidence has been clear-cut. A 2020 study published in the Journal of Economic Perspectives found that nearly 100 per cent of the Section 301 tariffs were passed through to US import prices. The aggregate annual cost to US households was estimated at US$1.4 billion per month, with the burden falling disproportionately on lower-income consumers who spend a larger share of disposable income on tradeable goods such as electronics, clothing, and furniture.

For China, the impact has been more diffuse. Headline real GDP growth has continued, but FDI inflows collapsed from US$189 billion in 2022 to just US$33 billion in 2023, reflecting both global investor caution and active divestment by multinational corporations restructuring supply chains under the "China plus one" strategy. Several Western firms, including Apple and Samsung, have relocated portions of their final assembly to Vietnam, India, and Mexico, while retaining design, components, and high-value intermediate-input production in China.

Source: Adapted from Financial Times, 6 February 2024 and Federal Reserve Bank of New York Liberty Street Economics, November 2023.

Question 1 **[2 marks]**

With reference to Table 2.1, calculate the percentage change in US imports from China between 2018 and 2023. Show your working.

Question 2 **[2 marks]**

Distinguish between "decoupling" and "re-routing" as used in Extract 2.1.

Question 3 **[4 marks]**

Using evidence from Extract 2.1, explain why the burden of the US tariffs on Chinese imports has fallen primarily on US consumers rather than Chinese producers.

Question 4 **[4 marks]**

With reference to Table 2.1, explain how the changes in US imports from Vietnam and Mexico between 2018 and 2023 illustrate the principle of trade diversion.

Question 5 **[6 marks]**

With the aid of an AD/AS diagram, explain how a sustained collapse in FDI inflows of the magnitude observed in China in 2023 could affect China's long-run economic growth.

Question 6 **[6 marks]**

Explain two reasons why the CHIPS and Science Act subsidies may not succeed in fully re-shoring semiconductor manufacturing to the United States.

Section Case Study 3. Japan — Exit from Deflation and the BOJ's Policy Normalisation

Table 3.1: Selected Japan Macroeconomic Indicators, 2019–2024

Indicator	2019	2020	2021	2022	2023	2024
Real GDP Growth (%)	−0.4	−4.2	2.7	1.0	1.9	0.1
Core CPI Inflation, ex-fresh food (%)	0.6	−0.2	−0.2	2.3	3.1	2.5
Spring Wage Settlement, "Shunto" (%)	2.07	1.90	1.78	2.07	3.58	5.28
BOJ Policy Rate (%, year-end)	−0.10	−0.10	−0.10	−0.10	−0.10	0.25
10-Year JGB Yield Ceiling (YCC, %)	none	0.25	0.25	0.50	1.00	abolished
JPY/USD Exchange Rate (year-end)	109	103	115	132	141	158
Tokyo Stock Price Index (TOPIX, year-end)	1721	1805	1992	1892	2366	2784

Source: Adapted from Bank of Japan, Japan Trade Union Confederation (Rengo), and Tokyo Stock Exchange.

Extract 3.1: Japan's wage-price virtuous cycle: long-awaited, fragile

For three decades after the bursting of its asset price bubble in 1991, Japan struggled with the developed world's most persistent deflationary problem. Core CPI inflation averaged just 0.1 per cent per year between 1995 and 2020. The Bank of Japan (BOJ) deployed an unprecedented toolkit in response: zero interest rates from 1999, quantitative easing from 2001, negative interest rates from 2016, and yield-curve control (YCC) from the same year, which committed the BOJ to capping the 10-year Japanese Government Bond (JGB) yield.

The post-pandemic inflation surge changed the picture decisively. Imported inflation from a weak yen and global commodity prices initially pushed core inflation above the BOJ's 2 per cent target in 2022, but the more important development came in 2023 and 2024. The Shunto spring wage round, in which Japan's largest labour federation Rengo negotiates with major employers, delivered increases of 3.58 per cent in 2023 and 5.28 per cent in 2024, the largest in more than three decades. "We are finally observing the wage-price virtuous cycle that has been missing for a generation," BOJ Governor Kazuo Ueda told reporters.

In March 2024, the BOJ ended negative interest rates and abolished YCC, citing high probability that the 2 per cent inflation target would be sustainably achieved. Yet the policy normalisation has been gradual and explicitly contingent. "If real wages remain negative or if global commodity prices fall sharply, we may revert to easing," Governor Ueda warned. The yen has continued to weaken, hitting 158 to the dollar in mid-2024, raising concerns that the exit from ultra-loose policy may be too slow to restore yen stability without further intervention.

A separate concern is fiscal: Japan's gross government debt is approximately 260 per cent of GDP, the highest among major advanced economies. Higher interest rates raise debt-servicing costs, and economists at the Ministry of Finance estimate that a one-percentage-point rise in average JGB yields would, over a decade, increase annual debt-service expenditure by approximately 3 trillion yen, roughly one-third of total defence spending.

Source: Adapted from Nikkei Asia, 19 March 2024 and Bank of Japan Outlook for Economic Activity and Prices, April 2024.

Question 1 **[2 marks]**

With reference to Table 3.1, describe the trend in Japan's core CPI inflation between 2019 and 2024.

Question 2 **[2 marks]**

Define "yield-curve control" as used in Extract 3.1.

Question 3 **[4 marks]**

Using evidence from Extract 3.1 and Table 3.1, explain why the BOJ regarded the 2023 and 2024 Shunto wage settlements as a critical signal for ending negative interest rates.

Question 4 **[4 marks]**

With reference to Table 3.1, explain how the depreciation of the yen from 109 per US dollar in 2019 to 158 per US dollar in 2024 would affect Japan's import prices and core inflation.

Question 5 **[6 marks]**

With the aid of an AD/AS diagram, explain how a sustained wage-price virtuous cycle could help Japan escape its long-run deflationary equilibrium.

Question 6 **[6 marks]**

Explain two reasons why Japan's high government debt-to-GDP ratio of approximately 260 per cent constrains the BOJ's ability to normalise monetary policy quickly. Refer to Extract 3.1 in your answer.

Section Case Study 4. Germany — Energy Shock, Fiscal Debt Brake, and the Industrial Recession

Table 4.1: Selected Germany Macroeconomic Indicators, 2019–2023

Indicator	2019	2020	2021	2022	2023
Real GDP Growth (%)	1.1	−3.8	3.2	1.9	−0.3
Industrial Production, annual change (%)	−3.7	−8.5	3.4	−1.0	−2.0
Producer Price Inflation (%)	1.1	−1.0	10.5	32.9	−2.4
CPI Inflation, harmonised (%)	1.4	0.4	3.2	8.7	6.0
Government Debt to GDP (%)	59.6	68.7	69.0	66.1	63.6
Natural Gas Import Volume (TWh)	1453	1392	1503	920	793
Manufacturing Employment (millions)	7.65	7.51	7.45	7.45	7.50

Source: Adapted from Destatis (Federal Statistical Office of Germany), Bundesbank, and European Commission AMECO database.

Extract 4.1: Germany's industrial recession and the debt-brake dilemma

Germany ended 2023 with the dubious distinction of being the only major economy to contract in real terms. The roots of the contraction stretched back to 2022, when the cessation of Russian natural gas supplies forced the rapid and expensive substitution to seaborne liquefied natural gas (LNG). German industry, particularly its energy-intensive chemicals, steel, and automotive sectors, faced electricity prices three to four times higher than US competitors. BASF, the world's largest chemical company, announced the permanent closure of two ammonia plants at its Ludwigshafen site and shifted growth investment to Louisiana.

The German government's response was complicated by a peculiar constitutional constraint. The "debt brake" (Schuldenbremse), adopted in 2009 in the wake of the global financial crisis, caps the federal structural deficit at 0.35 per cent of GDP except in cases of natural disaster or emergencies declared by the Bundestag. In November 2023, the Federal Constitutional Court invalidated the government's plan to redirect 60 billion euros of unused pandemic-emergency borrowing to a climate and transformation fund, ruling that emergency-classified debt cannot be repurposed across years. The decision left a hole in the 2024 federal budget of roughly 17 billion euros and forced the cabinet to delay several green-industrial subsidies and infrastructure projects.

The longer-term concern is that Germany's traditional growth model, based on cheap Russian gas, large exports to China, and a flexible labour market, has been simultaneously undermined on all three fronts. Russian gas is gone permanently. Chinese demand for high-end German engineering has slowed as China has moved up the value chain into electric vehicles, where domestic Chinese producers such as BYD now compete directly with German marques. And demographic decline has pushed manufacturing employment into structural decline despite tight labour markets in services. Economists at the Bundesbank revised their estimate of Germany's medium-term potential growth rate down from 1.2 per cent to 0.7 per cent, citing the combination of energy-cost re-pricing, weaker external demand, and a shrinking working-age population.

Source: Adapted from Bundesbank Monthly Report, January 2024 and The Economist, "Germany's economic woes", January 2024.

Question 1 **[2 marks]**

With reference to Table 4.1, describe what happened to Germany's natural gas import volume between 2021 and 2023.

Question 2 **[2 marks]**

Define "potential growth rate" and identify one piece of evidence from Extract 4.1 that suggests Germany's potential growth rate has fallen.

Question 3 **[4 marks]**

With reference to Table 4.1 and Extract 4.1, explain how the surge in producer price inflation from 1.1 per cent in 2019 to 32.9 per cent in 2022 was transmitted to headline CPI inflation.

Question 4 **[4 marks]**

Explain how the relocation of production by firms such as BASF, as described in Extract 4.1, would affect Germany's aggregate supply curve in the long run.

Question 5 **[6 marks]**

With the aid of an AD/AS diagram, explain how Germany's combination of energy cost shock and weakening external demand would generate the simultaneous contraction in real GDP and continued positive inflation observed in 2023.

Question 6 **[6 marks]**

Explain two reasons why Germany's debt-brake constraint may make the country more vulnerable to a prolonged recession than other major economies. Refer to Extract 4.1 in your answer.

End of Booklet — Set A